MSD MANUAL
Professional Version

Common Cold

(Upper Respiratory Infection; URI; Coryza)

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Reviewed/Revised May 2024 | Modified Apr 2025

The common cold is an acute, usually afebrile, self-limited viral infection causing upper respiratory symptoms, such as rhinorrhea, cough, and sore throat. The majority of cases are caused by rhinoviruses, but some are caused by coronaviruses, influenza viruses, or other viruses. Diagnosis is clinical. Handwashing helps prevent its spread. Treatment is supportive.

Approximately 50% of all colds are caused by one of the > 100 serotypes of rhinoviruses (1). Coronaviruses cause some outbreaks, and infections caused by influenza viruses, parainfluenza viruses, enteroviruses, adenoviruses, respiratory syncytial viruses, and metapneumoviruses may also manifest as the common cold, particularly in patients who are experiencing reinfection.

Rhinovirus infections are most common during fall and spring and are less common during winter. Rhinoviruses are most efficiently spread by direct person-to-person contact, although spread may also occur via large-particle aerosols.

The most potent deterrent to infection is the presence of specific neutralizing antibodies in the serum and secretions, induced by previous exposure to the same or a closely related virus. Susceptibility to colds is not affected by exposure to cold temperature, host health and nutrition, or upper respiratory tract abnormalities (eg, enlarged tonsils or adenoids).

Reference

1. Mäkelä MJ, Puhakka T, Ruuskanen O, et al: Viruses and bacteria in the etiology of the common cold. *J Clin Microbiol*. 1998;36(2):539-542. doi:10.1128/JCM.36.2.539-542.1998

Symptoms and Signs of Common Cold

After an incubation period of 24 to 72 hours, cold symptoms begin with a scratchy or sore throat, followed by sneezing, rhinorrhea, nasal obstruction, and malaise. Temperature is usually normal, particularly when the pathogen is a rhinovirus or coronavirus. Nasal secretions are watery and profuse during the first days but then become more mucoid and purulent. Mucopurulent secretions do not

indicate a bacterial superinfection. Cough is usually mild but often lasts into the second week. Most symptoms due to uncomplicated colds resolve within 10 days.

Colds may exacerbate asthma and chronic bronchitis.

Purulent sputum or significant lower respiratory tract symptoms are unusual with rhinovirus infection. Purulent sinusitis and otitis media may result from the viral infection itself or from secondary bacterial infection.

Diagnosis of Common Cold

- History and physical examination

Diagnosis of the common cold is generally made clinically and presumptively, without diagnostic tests, although polymerase chain reaction (PCR) testing is available in many multiplex platforms. Consider testing in patients for whom knowing the specific pathogen will affect clinical management, such as when specific antiviral therapy is indicated (eg, suspected influenza, COVID-19, or RSV).

Allergic rhinitis is the most important consideration in differential diagnosis.

Treatment of Common Cold

- Symptomatic treatment

Treatment is supportive. Antipyretics and analgesics may relieve fever and sore throat.

Nasal decongestants may reduce nasal obstruction. Topical nasal decongestants are more effective than oral decongestants, but the use of topical medications for > 3 to 5 days may result in rebound congestion.

Rhinorrhea may be relieved with first-generation antihistamines (eg, chlorpheniramine) or intranasal ipratropium bromide (2 sprays of a 0.03% solution 2 or 3 times a day); however, these medications offer minimal benefit and should be used with caution in older adults, people with benign prostatic hypertrophy, or angle-closure glaucoma. First-generation antihistamines frequently cause sedation, but second-generation (nonsedating) antihistamines are ineffective for treating the common cold.

Antihistamines and decongestants are not recommended for children < 4 years.

Zinc (1), echinacea (2), and vitamin C (3) have all been evaluated as common cold therapies, but none has been clearly shown to be beneficial. Although zinc lozenges may reduce the duration of common cold symptoms, that benefit must be balanced against adverse effects, notably a bad taste and nausea (1). Importantly, there is concern about permanent anosmia with use of intranasal zine (4).

Antibiotics should not be given unless there is clear evidence of secondary bacterial infection.

Treatment references

1. Singh M, Das RR. Zinc for the common cold. *Cochrane Database Syst Rev*. 2013;(6):CD001364. Published 2013 Jun 18. doi:10.1002/14651858.CD001364.pub4
2. Karsch-Völk M, Barrett B, Kiefer D, Bauer R, Ardjomand-Woelkart K, Linde K: Echinacea for preventing and treating the common cold. *Cochrane Database Syst Rev*. 2014;2014(2):CD000530. Published 2014 Feb 20. doi:10.1002/14651858.CD000530.pub3
3. Hemilä H, Chalker E: Vitamin C for preventing and treating the common cold. *Cochrane Database Syst Rev*. 2013;2013(1):CD000980. Published 2013 Jan 31. doi:10.1002/14651858.CD000980.pub4
4. D'Cruze H, Arroll B, Kenealy T: Is intranasal zinc effective and safe for the common cold? A systematic review and meta-analysis. *J Prim Health Care*. 2009;1(2):134-139.

Prevention of Common Cold

There are no vaccines for the common cold.

Measures to prevent spread of the common cold include handwashing, avoiding close contact with people with a cold, covering the mouth and nose when coughing or sneezing, and use of surface disinfectant in a contaminated environment.

Key Points

- **Many viruses can cause the common cold; rhinoviruses cause approximately 50% of colds.**
- **Susceptibility to colds is not affected by exposure to cold, host health and nutrition, or the presence of upper respiratory tract abnormalities.**
- **Antihistamines may be used to relieve rhinorrhea, but they offer minimal benefit and should be used with caution in older adults and avoided in children < 4 years.**
- **Topical and oral decongestants relieve nasal obstruction, but repeated use may cause rebound congestion.**
- **Many substances have been evaluated for prevention and treatment, but none has clearly been shown to be beneficial.**

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